Seminar 1 · Preferences and utility

Microeconomics

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Relations, contours, representations and counterexamples.

The route

1. Classify relations before writing utility.
2. Defend a true claim; break a false one.
3. Separate utility curvature from preference geometry.
4. Distinguish one purchase from the full choice set.
5. Put assumptions back into theorems.

What counts as an answer?

“False, because the preferences are strange” is not a counterexample.

- State the domain and the relation or function.
- Name the bundles, sequence or menu.
- Check the hypothesis.
- Exhibit the precise failed inequality or implication.

1A · Coordinatewise comparison

On $X = \mathbb{R}^2_+$,

$$x \succeq_A y \iff x_1 \geq y_1 \text{ and } x_2 \geq y_2.$$

Complete? Transitive? Continuous?

Incomplete: $(1, 0)$ and $(0, 1)$.

Transitive: chain the two coordinate inequalities.

Continuous: two weak inequalities survive limits.

1B · Replace AND by OR

NES

$$x \succeq_B y \iff x_1 \geq y_1 \text{ or } x_2 \geq y_2.$$

Complete? Transitive? Continuous?

Complete: failing both inequalities makes the reverse comparison true.

Intransitive: $x = (1, 0)$, $y = (0, 2)$, $z = (2, 1)$ satisfy $x \succeq_B y \succeq_B z$ but $x \not\succeq_B z$.

Continuous: its graph is a finite union of closed sets.

1C · A tiny tie-breaker?

Compare $x_1 + x_2$ first; when totals tie, compare x_1.

Can $x_1 + x_2 + \varepsilon x_1$, for some fixed $\varepsilon > 0$, represent this ranking?
No. Take $a = (0, 1 + \delta)$, $b = (1, 0)$ and $0 < \delta < \varepsilon$.
The primary total ranks a higher; the weighted sum ranks b higher.
The relation is complete and transitive, but discontinuous.

1C · Find the broken limit

Find $x^n \succeq_C y^n$ whose limits reverse the comparison.

$$x^n = (0, 1 + 1/n), \qquad y^n = (1, 0).$$

Every x^n has the larger total. At the limit, totals tie and $(1, 0)$ wins the tie-breaker.

2 · True or false?

1. Incomparability means indifference.
2. Utility representation implies rationality.
3. Discontinuous utility implies discontinuous preferences.

1. False: neither comparison versus both comparisons.
2. True: inherit completeness and transitivity from $\geq$.
3. False: $x_1 + \mathbf{1}\{x_1 \geq 1\}$ represents the same continuous ranking as x_1.

2 · Three more claims

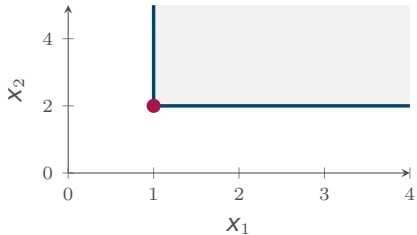
1. Convex utility implies convex preferences.
2. Increasing transformations preserve convex preferences, but need not preserve concavity of utility.
3. LNS implies utility is unbounded above.

1. False: $u = x_1^2 + x_2^2$; mix $(2, 0)$ and $(0, 2)$.
2. True: same rankings and upper contour sets.
3. False: $u = 1 - e^{-(x_1 + x_2)}$ is bounded and LNS.

3 · A kink is not a pathology

$$u(x) = \min\{2x_1, x_2\}, \qquad X = \mathbb{R}^2_+.$$

Draw the indifference set and the upper contour set through $(1, 2)$.



$u = 2$: two rays.
$u \geq 2$: $x_1 \geq 1$ and $x_2 \geq 2$.

3 · Classify the preferences

Rational? Continuous? Convex? Strictly convex? Monotone? Strongly monotone? LNS?

Rational, continuous and convex.

Not strictly convex: $(1, 3) \sim (1, 4)$ and their midpoint.

Monotone and LNS: increase both goods slightly.

Not strongly monotone: $(1, 2) \sim (2, 2)$.

3 · Transform the index

$$v = e^u, \qquad w = \mathbf{1}\{u \geq 2\}.$$

Same preferences? Concave utility?

u is concave: the minimum of these two linear functions is concave.

v preserves preferences but is not concave: along $(t, 2t)$ it equals e^{2t}.

w loses strict comparisons, e.g. between $(1, 2)$ and $(2, 4)$. It is not another representation.

4 · A choice changes when the menu expands

NES

Selected a from $\{a, b\}$; selected b from $\{a, b, c\}$.

Does this refute a fixed rational ranking?

Not if each record is only one selected maximiser.

$a \sim b \succ c$ permits both selections.

We need to know whether unselected alternatives were also best.

4 · Now these are full best-alternative sets

NES

$$C(\{a,b\}) = \{a\}, \qquad C(\{a,b,c\}) = \{b\}.$$

Can a rational ranking generate these correspondences?

No. The pair implies $a \succ b$, whereas the triple implies $b \succ a$.

The interpretation of the data changes the conclusion.

5 · An optimum cannot leave slack under LNS

$$B = \{x \geq 0 : p \cdot x \leq m\}, \qquad p \gg 0,\ m > 0.$$

Assume a best bundle x^* exists.

Prove budget exhaustion without differentiating.

If $s = m - p \cdot x^* > 0$, choose $\delta = s/(2\|p\|)$.

Every consumption bundle within distance δ remains affordable.

LNS gives a strictly better nearby bundle: contradiction.

5 · Does the converse hold?

$$u = -[(x_1 - 2)^2 + (x_2 - 2)^2], \quad p = (1, 1), \quad m = 1.$$

Every optimum exhausts this budget. Does that prove LNS?

No. Below this budget line, slightly increasing either good improves utility.

But LNS fails at $(2, 2)$, the global bliss point.

A statement about one budget cannot establish a global property.

5 · A failed counterexample is useful

NES

On $x_1 + x_2 \leq 1$, lexicographically maximise x_1, then minimise x_2.

Does this show that LNS need not give existence?

No: $(1, 0)$ is a maximiser.

A valid repair uses $X = [0, 1]$, $r(x) = x$ for $x < 1$, $r(1) = -1$.

It is LNS but has no maximum, despite compactness. Continuity fails.

6 · Reserve: discretise lexicographic preferences

NES

$$X_N = \{0, 1/N, \dots, 1\}^2, \qquad U_N = Mx_1 + x_2.$$

Which M works? Why does the same trick fail on the square?

$M > N$: the smallest primary gain outweighs the largest secondary loss.

On $[0,1]^2$, primary gains can be arbitrarily small.

In fact no real-valued representation exists there: use disjoint utility intervals and countability of the rationals.

What to carry into the homework

- Quantifiers and domains are part of every claim.
- A representation is ordinal; a curvature property need not be.
- Convex preferences mean convex *upper* contour sets.
- Keep full best sets separate from selected purchases.
- Existence, uniqueness and budget exhaustion need different assumptions.